Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/14/2017
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
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No. of Post Docs:	4	**No. of PhD Degrees:**	12
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2017: Key members are: Sandhya Bhatnagar, Life Sciences Division, LBNL; Halina E. Witkowska, University of California, San Francisco; Bernard Rabin, University of Maryland, Baltimore Campus; Richard Britten, Eastern Virginia Medical School; Tore Straume, NASA Ames, Space Biosciences Division; Leif E. Peterson, Houston Methodist Research Institute. Changes in Personnel, as of February 2016: Dr. Witkowska has formally retired. She is actively helping us to prepare manuscripts. We have initiated a collaboration with EMSL at PNNL to perform the mulit-omic analyses for metabolites and protein changes in the tissue collected in our experiments.		
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Task Description:	A major goal of the NASA space radiation research program is to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. Our research plan builds on the mounting evidence of the radiation-sensitivity of the hippocampus, which is important for memory and performance, and the choroid plexus (CP), which produces cerebral spinal fluid and maintains the blood-brain barrier. Using transcriptomic, proteomic, metabolomics, and immunohistochemical analyses of CNS tissue from rodents exposed to simulated space radiation, we have identified molecular and cellular changes associated with oxidative stress, neuro-inflammation, vasculature damage, changes in the protein composition of cerebral spinal fluid, and amyloid deposits such as those found in Alzheimer disease (AD). We propose new research to investigate the low-fluence responses and post-exposure-time dependence of these findings, to identify the molecular mechanisms of inter-individual variations in behavioral responses after radiation exposure, and to validate the predicted mechanisms using tissue of irradiated and behaviorally characterized rodents. Our specific aims are to: (1) Investigate the cellular and functional abnormalities in the hippocampus and CP of HZE-irradiated rats that are predicted by our multi-omics findings with the emphasis on defects associated with AD. (2) Investigate hippocampal and CP radiation damage responses associated with inter-individual variation in radiation-induced neurocognitive deficits, using proteomics, RNA expression, metabolomics, and in situ CNS analyses, with emphasis on tissue defects associated with AD. (3) Characterize the radiation sensitivity of persistent CP epithelial and endothelial damage in AD mouse models and examine their associations with neuro-inflammation and other defects associated with AD. This research is foundational to establish quantitative mechanism-based predictors for neuro-cognitive risks in individuals exposed to high-LET (linear energy transfer) space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is designed to provide NASA with a mechanistic understanding of the risks for impaired neurocognitive performance and individual susceptibility for neurological diseases after exposure to high energy heavy ions (HZE) radiation, with emphasis on persistent neurological risks. Our approach: (a) employs state-of-the-art multi-omics and tissue technologies to investigate CNS damage response functions in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neurocognitive performance results, and (c) validates the predicted damage response functions in brain tissue. Our project focuses on radiation damage to the hippocampus (site of adult neurogenesis; spatial and performance memories) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research strategy distinguishes between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) inter-individual variations in radiation-induced CNS risks for neurocognitive performance and neurological diseases, i.e., Alzheimer Disease (AD). Our research is conducted in outbred strains of rats to capture the effects of genetic variation and in genetically defined double and triple mutation AD mice. Successful completion of this project will provide a mechanistic foundation for space-radiation-induced CNS risk that is based on quantitative molecular and cellular defects and their associations with measured neurocognitive and behavioral abnormalities. The project considers the time of onset and the persistence of effects, as well as inter-individual differences in responses. The project will employ space-relevant fluences of simulated HZE radiation, ensuring relevance for astronaut risk assessment. The results are expected to play a key role in helping to reduce the uncertainty associated with predicting the CNS risk for deep space exploration missions. This research is also designed to contribute to the identification of molecular targets for biological countermeasures to reduce CNS risks from space radiation.
	Our research goal is to understand the molecular mechanisms of individual variations in CNS tissue damage, behavioral performance, and risks for neurological diseases after exposure to HZE space radiation, using the Sprague-Dawley rat model. The animals are exposed to various doses of 1GeV/n 56Fe at the NASA Space Radiation Laboratory (NSRL), and tested for memory, operant behavior, movement, and anxiety. Tissues for multi-omic profiling (proteomic, transcriptomic, and metabolomics) from cerebral spinal fluid (CSF), choroid plexus (CP), cortex, and hippocampus were collected between 3 days to 22 months (m) after exposure to generate tissue damage hypotheses that are tested under an experimental protocol that allows for direct comparison of behavior performance, multi-omics predictions, and histopathological validation within the same animal. Genetic mouse models are applied to validate the multi-omic and histological findings related to Alzheimer disease. Persistent abnormalities in 'omic expression predicted choroid plexus histopathologies, vascular abnormalities, and amyloid deposition that were confirmed in tissues from irradiated animals. Exposures as low as 10cGy 56Fe produced abnormalities in CSF protein and tissue transcript expression at 2 and 22 m after exposure, respectively. Affected genes were associated with neurological diseases, vasculature defects, and amyloid/fibril abnormalities, and these predictions were tested in tissue sections of the same animals. Beginning at ~4 m and increasing at 9 and 22 m after exposure, irradiated brains showed increased CP endothelial vascular fibrosis and reduced epithelial expression of transthyretin (TTR), a CSF protein associated with ß-amyloid clearance. Irradiated rats also showed increased frequencies of amyloids (Congo Red), extracellular aggregations of amyloid beta A4, and increased extracellular TTR amyloid bodies throughout the CNS at 9, 15, and 22 m after exposure. Our findings suggest that CP and CSF are early targets of radiation-induced abnormalities in gene expression that are associated with risks for late-onset CNS histopathology. Characterization of behavioral performance of susceptible and resistant animals exposed to space radiation. Spatial memory learning: We investigated the nature of dose responses for spatial memory learning among genetically outbred male Wistar rats exposed to graded doses of 56Fe particles (sham, 5, 10, 15, and 20 cGy; 1 GeV/n). Spatial memory learning was assessed at 3 months after exposure on a Barnes maze. Radiation induced a monotonic decline in learning across these doses following a linear response and fell into three performance categories: (1) progressively impaired learning between sham and 10 cGy exposures, (2) no detectable learning between 10 and 15 cGy exposures, and (3) worsening after 15 and 20 cGy exposures. The proportions of poor learners and the magnitude of their impairment also followed linear responses with doubling doses of ~10 cGy. In contrast, good learners showed no detectable learning deficits in this dose range. Our findings suggest that genetically diverse individuals can vary substantially in spatial memory learning and that exposures to low-dose exposures appear to predominantly impact poor performers. This hypothesis invites future investigations of the underlying physiologic and genetic mechanisms of

Task Progress:	inter-individual variations in brain function after low-dose HZE radiation exposures. Analyses of dose response showed that animals who are poor performers for spatial memory learning on the Barnes maze showed a decline in learning performance that was linear with a doubling dose of ~15cGy, while high performance learners show no detectable radiation effect at doses up to 20 cGy. Comparative behavioral responses to low-dose 56Fe exposures on three behavioral platforms: 124 male outbred Sprague Dawley rats were irradiated at NSRL (56Fe; 1GeV/n; sham, 10-20cGy; n=40-44; 2 m old at exposure) and evaluated for behavior by video analyses on open field (OF, pre, and 3 m post), elevated plus maze (EPM, 1 m post) and Barnes maze (BM, 3 m post; 6 trials over 3 days). A reference cohort was exposed to 50 or 200 cGy of X-rays and evaluated by EPM before and 1-m post exposure. Preliminary findings are: Movement parameters: 10 and 20 cGy 56Fe showed no differential effect on movement in OF, EPM, and BM, e.g., distance traveled. In contrast, 50 cGy of X-rays decreased movement on the EPM. EPM parameters: 10 and 20 cGy 56Fe showed no effect on open-arm and closed-arm endpoints. In contrast, rats exposed to 50 cGy X-ray showed fewer entries into the arena center and reduced closed arm activity. OF parameters: 10 and 20 cGy 56Fe animals spent less time in the arena center and more time at borders. BM endpoints: 10 and 20 cGy 56Fe decreased the escape latency (EL) during the first 3 trials, but not in trials 4-6. Radiation did not change head-poke frequencies or REL3 (i.e., ratio of days 3 to 1 EL). Our three-behavior study indicates that low-dose exposures of 10 and 20 cGy 56Fe to 2-m old male Sprague Dawley rats induced persistent minor changes in anxiety-related endpoints with no detectable effects on animal movement. Our findings that behavioral responses are variable suggest that individual responses may be affected by husbandry, age, physiology and/or genetics. Molecular characterization of animal-to-animal variations in anxiety behaviors. We compared proteomic profiles of tissues from the choroid plexus and hippocampus of anxiety-characterized animals exposed to 56Fe particles. Male Sprague-Dawley rats, exposed at 6 months of age to two doses of 56Fe particles (1 GeV/n; sham, 10cGy and 100cGy; n=24 per group), were evaluated ~1 m after exposure on EPM. Radiation exposure significantly increased closed-arm preference in each dose group. The proteomic profiles of CP and hippocampal tissues collected from animals at the extremes of the anxiety distributions in each dose group were compared by RP-HPLC and LC/MS: 1448 choroid plexus proteins and 1209 hippocampal proteins were identified. Proteins showing significant correlations with baseline anxiety, independent of exposure, were significantly associated with upstream regulation by MAPT, PSEN1, APP, and HTT, which are associated with Alzheimer disease and other neurological abnormalities. In contrast, proteins showing significant correlation with anxiety only in irradiated animals were significantly associated with upstream regulation by TP53 and MYC, which are two radiation damage response proteins. The anxiety discriminating proteins are associated with neuro-inflammation, neurotangle and amyloid biology, certain neurotransmitters, and synaptic activity. A confirmatory experiment with expanded behavior testing, multi-tissue multi-omics, and additional in situ endpoints is in progress. Working model of the tissue mechanisms that predict individual variation in behavioral deficits and risks for late-onset neurological abnormalities in irradiated animals. We are building a mechanistic model of individual variation in CNS risks for abnormal behavior and late-onset neurological diseases. This model is based on our observations that the CP and CSF, which are major components of the blood-CSF brain barrier, are sensitive to radiation-induced molecular abnormalities that persist long after exposure, in some cases for the life of the animal. The subset of molecular abnormalities in expression that persist long after exposure predict subsequent tissue damage associated with late-onset neurological abnormalities, such as regional cellular histopathologies, vascular abnormalities, and the formation of protein aggregates and amyloids. Our model also incorporates the observation that inter-individual variations in behavior outcomes are correlated with molecular profile changes in CNS subregions, offering a promising approach to identify individuals that may be more susceptible to low-dose HZE exposures. In our model, individual susceptibility to radiation induced anxiety responses and memory deficits are associated with the differential expression of CP and hippocampal proteins, which provides a novel window into understanding molecular mechanisms that may predict individual variations in CNS responses to space radiation exposures and may lead to the future development of mechanism-based molecular countermeasures.
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